

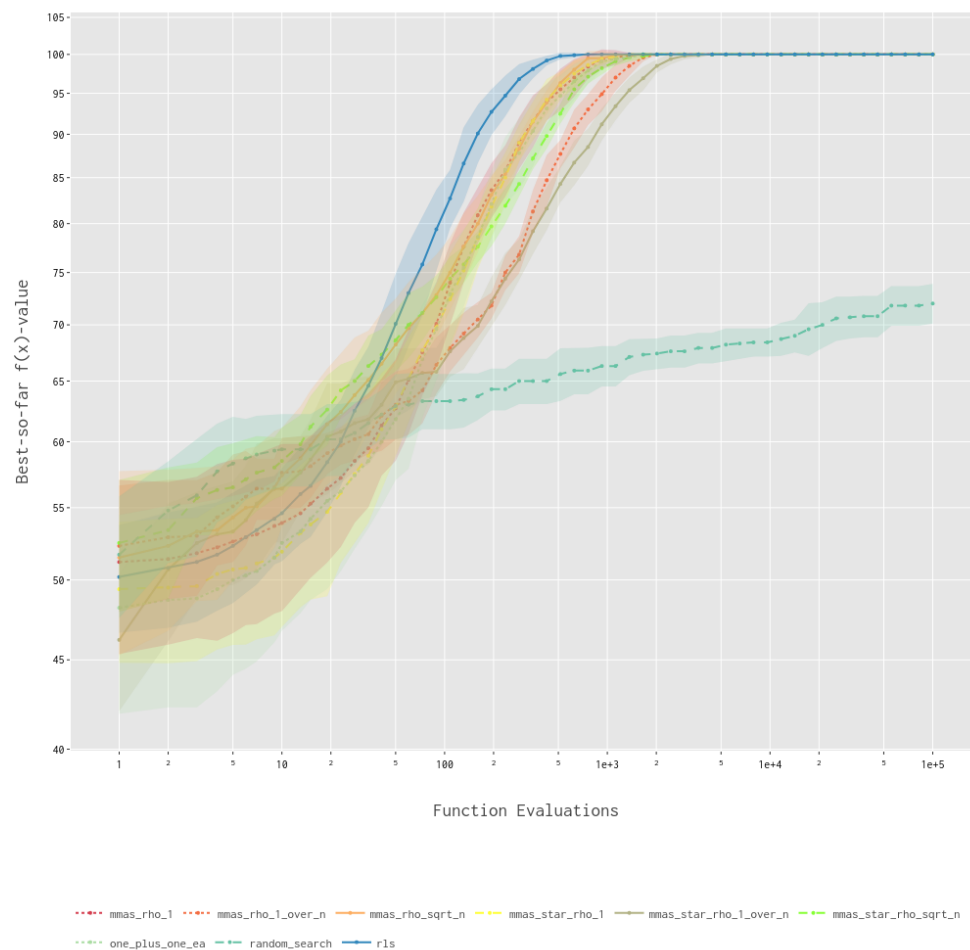
Exercise 4: MMAS and MMAS* Analysis

1. Introduction

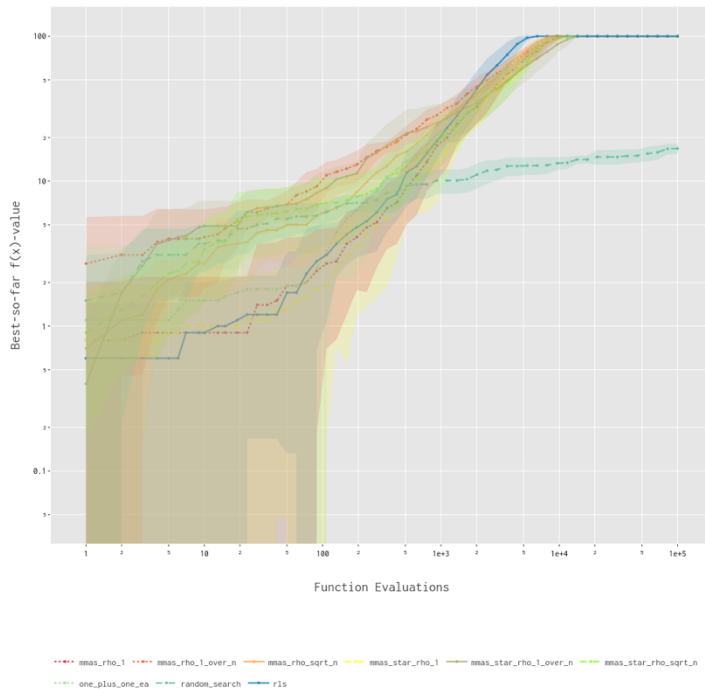
This report compares two ACO algorithms: MMAS and MMAS* on seven benchmark functions. The key difference: MMAS accepts equal fitness solutions ($f(x) \geq f(x^*)$) while MMAS* requires strict improvements ($f(x) > f(x^*)$). We tested three evaporation rates: $\rho \in \{1, 1/\sqrt{n}, 1/n\}$ where $n=100$, running each configuration 10 times for 100,000 evaluations. Results were compared against Random Search, RLS, and (1+1) EA from Exercise 2.

2. Results by Function

2.1 F1 - OneMax



RLS dominated, reaching optimum around 500 evaluations. Among ACO variants, $\rho=1/\sqrt{n}$ performed best but remained slower (~2,000-3,000 evaluations). Low evaporation ($\rho=1/n$) was extremely slow, only catching up around 50,000 evaluations. Random Search plateaued at ~72/100. MMAS and MMAS* showed identical performance, as expected for unimodal functions.

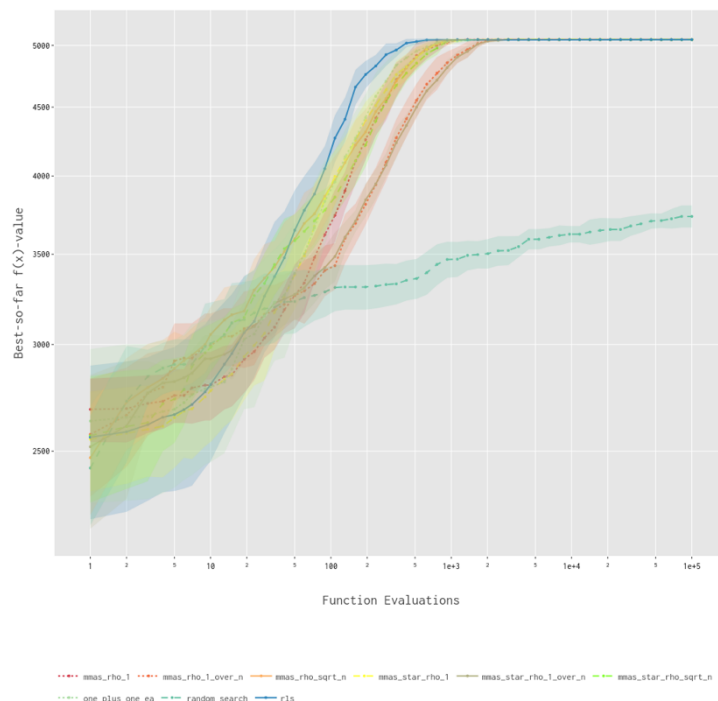


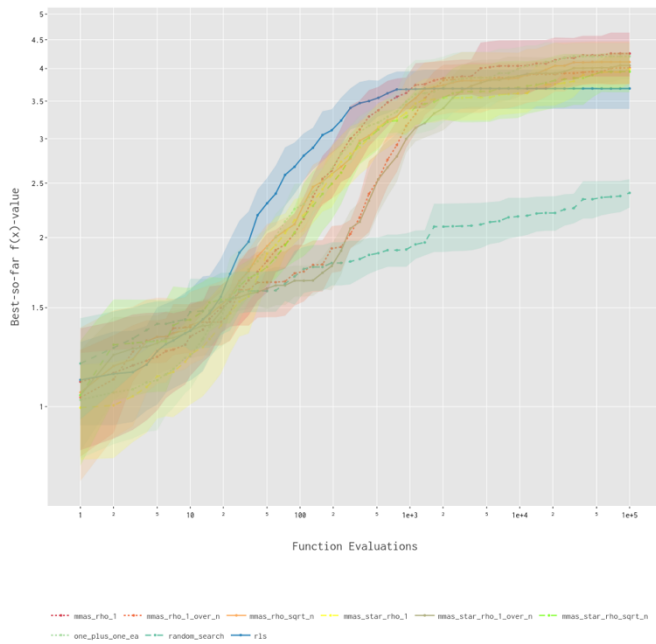
2.2 F2 - LeadingOnes

RLS excelled on this sequential function ($\sim 10,000$ evaluations to optimum). ACO algorithms struggled more than on OneMax due to prefix dependencies. Best ACO ($\rho=1/\sqrt{n}$) required 5,000–8,000 evaluations. Random Search performed poorly ($\sim 17/100$). Again, no difference between MMAS and MMAS*.

2.3 F3 - Linear

Similar pattern to OneMax. RLS fastest, followed by $(1+1)$ EA, then ACO with $\rho=1/\sqrt{n}$. Random Search achieved $\sim 3,700/5,050$. Higher variance visible for ACO algorithms. MMAS and MMAS* remained indistinguishable.



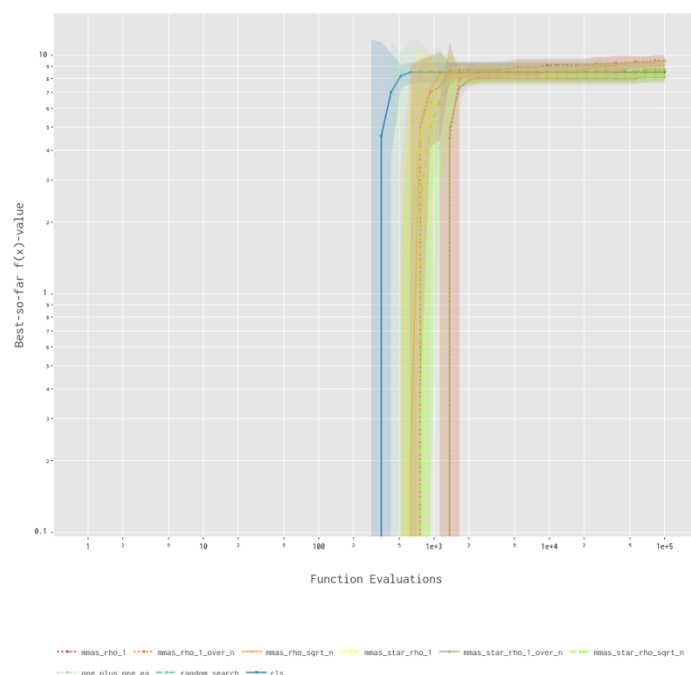


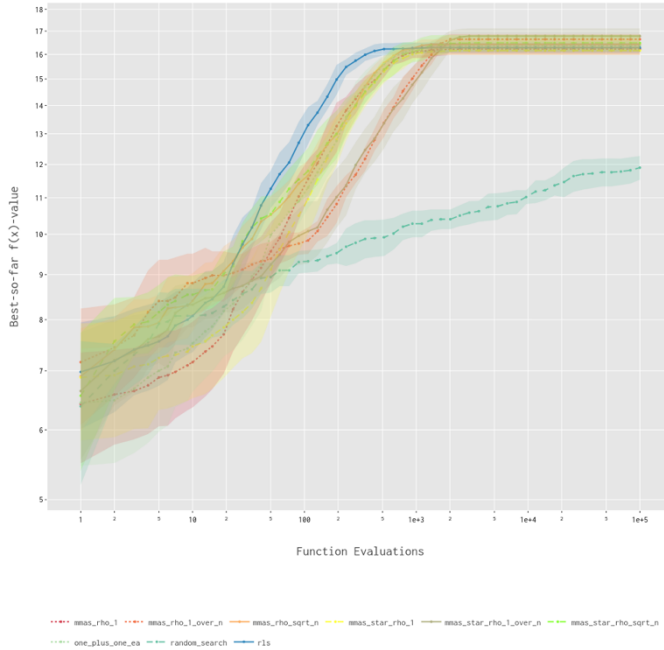
2.4 F18 - LABS

First function where ACO became competitive. MMAS with $\rho=1$ matched RLS performance (~ 3.7 - 3.8 fitness). Both $\rho=1$ and $\rho=1/\sqrt{n}$ worked well, unlike unimodal functions. Wide confidence bands indicate problem difficulty. (1+1) EA performed worse than some ACO variants—first clear ACO advantage.

2.5 F23 – Nqueens

RLS showed sharp improvement around 300 evaluations, plateauing at ~ 8 - 9 queens. All other algorithms clustered together at similar performance. Critically, this plateau function showed **no observable difference between MMAS and MMAS***, contradicting theoretical predictions that MMAS should outperform MMAS* on plateaus.



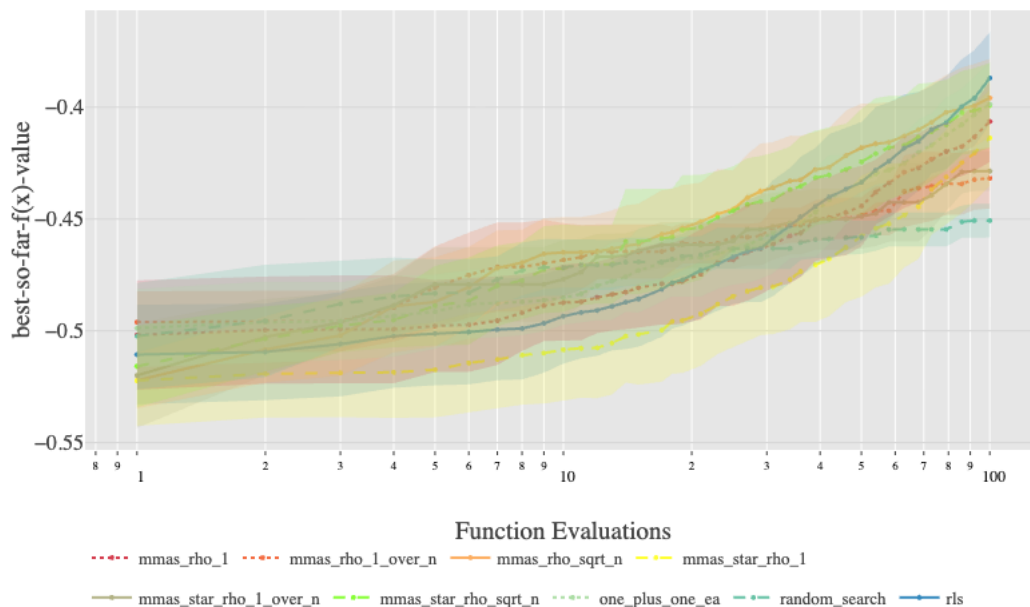


2.6 F24 - ConcatenatedTrap

RLS best (~ 16.5), MMAS $\rho=1/\sqrt{n}$ close second. Large variance across runs indicates deceptive landscape difficulty. Random Search relatively better (~ 12) since it's not misled by deceptive gradient. No MMAS vs MMAS* difference observed.

2.7 F25 – NKLandscapes

ACO algorithms clearly outperformed evolutionary baselines. MMAS* $\rho=1/\sqrt{n}$ achieved best results (~ -0.40), with MMAS performing identically. (1+1) EA fell to middle of pack. Despite being a plateau function with epistasis, *MMAS and MMAS* showed no performance difference**.



3. Analysis

3.1 Effect of ρ Parameter

$\rho = 1/\sqrt{n}$: Won or tied on all 7 functions. Optimal balance between adaptation and persistence. Recommended default.

$\rho = 1$: Competitive on complex landscapes (F18, F24). Behaves like (1+1) EA when pheromones hit boundaries quickly.

$\rho = 1/n$: Consistently worst. Too slow within 100k budget. Should be avoided.

3.2 MMAS vs MMAS*

Expected: MMAS should outperform MMAS* on plateaus (F23, F25) by accepting equal-fitness solutions.

Observed: No measurable difference on any function, including plateaus. Curves overlay exactly for matching ρ values.

Possible explanations:

- 100k evaluation budget may be insufficient for the differences to emerge.
- Benchmark plateaus more complex than the theoretical examples (NEEDLE).
- Pheromone update mechanism dominates over the selection mechanism.
- Theory considers asymptotic behaviour, not finite-time performance.

3.3 ACO vs Evolutionary Algorithms

RLS/EA win: Unimodal functions (F1, F2, F3) and constrained problems (F23) ACO competitive/wins: Complex landscapes (F18, F24, F25)

Pattern: ACO's pheromone memory helps on rugged multi-modal landscapes but adds unnecessary overhead on smooth functions.

4. Conclusions

Key findings:

1. $\rho = 1/\sqrt{n}$ is robust across all problem types
2. MMAS vs MMAS* distinction has negligible practical impact
3. Parameter choice (ρ) matters far more than selection mechanism
4. ACO excels on complex landscapes, struggles on simple unimodal functions

Practical recommendations:

- Default to $\rho = 1/\sqrt{n}$
- Use RLS for known unimodal functions

- Either MMAS or MMAS* acceptable—no meaningful difference observed
- Avoid $\rho = 1/n$

The surprising lack of MMAS/MMAS* differences suggest theoretical worst-case analyses may not capture typical behaviour on practical benchmarks. Parameter tuning (ρ) proves more important than algorithmic details (selection mechanism) for practitioners.